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Roll No. | | | | | | | |

Paper Code: TCE 813

B.Tech (Civil Engineering)
End Semester Examination 2024
VIII *Semester*
Hydropower Engineering

Time: Three Hours

MM: 100

Note:

(i) All questions are compulsory.
(ii) Answer any two sub questions in each main question.
(iii) Total marks for each main question is **Twenty**

Q1

(10X 2 = 20 Marks)

a) The yearly output of a base load plant with 25 MW installed capacity is 125×10^{6} kWh. The plant takes a peak of 22.5 MW. Calculate the annual load factor and capacity factor?

b) A power development site has the average estimated river flows as given in the table below (flow is million m^3).

Months	Jan	Feb	Mar	April	May	Jun	Jul	Aug	Sept	Oct	Nov	Dec
Flow	68	35	77	53	15	17	35	89	92	44	54	87

Calculate the required capacity of the reservoir in order to produce 14500 kW continuously throughout the year at an overall efficiency of 85% under a head of 110 m. CO1

c) Predict the load of a town in year 2023 using Scheer formula and the following data:
Per capita load in year 2018 = 350 kWh
Population growth rate = 7.5 % (constant for the period)
Population in year 2018 = 90,00000

Q2

(10X 2 = 20 Marks)

a) Run-of-river plant on a stream has inflow of 20 Cumecs and net head of 30m with provision for pondage to meet daily peak demand with a load factor of 60%. Calculate
a) The power generation capacity of the plant at 80% overall efficiency,
b) The plant runs as peaking station for 3 hrs and balance period in the day for average load. Estimate the amount of pondage needed?

b) In a hydroelectric station, water is available at the rate of 175m3/s under a head of 18m the turbine run at speed of 150 r.p.m with overall efficiency of 82%. Calculate the number of turbine required if they have the maximum specific speed of 460 CO2

c) Explain run-of-river plants also describe the general arrangements of the run-of-river plants?

Q3

(10X 2 = 20 Marks)

a) Describe all the ways of locating an underground power station? Explain with neat sketch?

b) "Some of the power plants adopt two other types of super-structure in which the machine hall concept does not exist" Explain the types of super-structure? CO3

c) Describe all the three main divisions of a power station with a neat sketch?

Q4 (10X 2 = 20 Marks)

a) Determine the expressions for the water hammer pressure in case of a rigid pipe and in case of an elastic pipe?

b) A hydro-electric station is not to be supplied with 10 m^3/sec of water through a penstock which has a friction factor of 0.016. The maximum normal head of the penstock is 50 kg/cm^2 and a water hammer overpressure of 20% over the normal pressure is anticipated. The safe head in the steel used is presumed to be 3000 kg/cm^2. The ready penstocks at site are likely to cost Rs 15000 per tonne including erection charges. The life of the project is 50 years and the rate of interest is 7 per cent. It is proposed to sell the energy at the rate of Re 0.03/ kWh. Predict the optimum diameter of the penstock, given the OMR costs are 5 per cent. Assume the turbine efficiency to 90 per cent and the annual load factor is 0.4? CO4

c) Determine the economic diameter of the penstock for a power plant having a rated head of 65.532 m discharge 215.208 cumecs, rated HP is 85000, there are two identical turbines fed by the penstock?

Q5 (10X 2 = 20 Marks)

a) Explain the layout arrangement and the hydraulics of turbines?

b) A Francis turbine on a major hydro-power project in Pakistan is to operate under a net head of 150 m. The desired output per turbine is 175 MW and the synchronous generator speed is 136.4 rpm. Model studies indicate an overall efficiency of 91.5 percent and the prototype efficiency is likely to be at least 0.5 percent. Hyd. Efficiency is estimated as 96 percent. Calculate the specific speed and the discharge to the machine. Also suggest suitable values of exist angle of guide vanes, the vane angles of the wheel , the inlet and outlet diameter of the wheel and the breadth of the wheel at the inlet. Make suitable assumptions. CO5

c) At the Volga imeni V.I. Lenin hydroelectric plant, in USSR,the Kaplan turbine used has the following data: Operation head is 22.5m, Output power at this head is 126 MW, Discharge at this head is 615 m^3/sec, Speed is 68.2 rpm, Runner tip-to-tip diameter (D) is 9.3 m, Hub diameter (D_h) is 4.3 m, Number of blades is 6. Calculate the speed ratio, the flow ratio and the overall efficiency and the maximum suction draft head.

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Roll No.

End Semester Examination 2024

Name of the Course: B. Tech (Civil Engineering)
Name of the Paper: Earthquake Resistant Design of Buildings
Time: 3 Hours

Semester: VIII
Paper Code: TCE801
Maximum Marks: 100

Note:

All questions are compulsory
Answer any two sub questions among a, b & c in each main questions
Total marks in each question are twenty
Each question carries 10 marks

Q1 (20 marks)

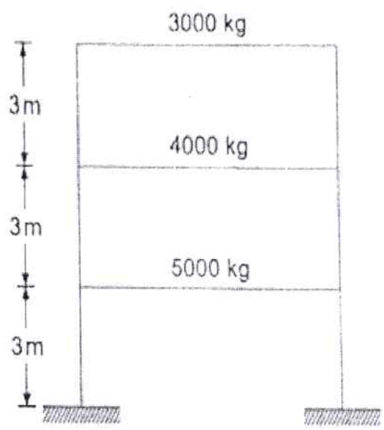
(a) Derivation of natural frequency by energy method.

(b) Explain response spectrum with the help of diagram

(c) Explain method of seismic analysis for SDOF system **CO1**

Q2 (20 marks)

(a) Determine the natural frequencies and the mode shapes for the shear building as shown in the fig. below. Take $E_L = 4.5 \times 10^6$ N-mm^2 for all columns. **CO2**



(b) Explain the two degree of freedom systems and MDOF.

(c) Explain the damping and its types.

Q3 (20 marks)

(a) Explain the interior structure of the earth

(b) Discuss the elastic rebound theory **CO3**

(c) Discuss the plate tectonic theory

Q4 (20 marks)

(a) What do you understand by inelastic seismic response?

(b) Explain the Dynamic Response of Spectrum representation for elastic systems **CO4**

(c) Write a brief note about soil structure interaction.

Q5 (20 marks)

(a) Write short notes on: **CO5**

   a. Design basis earthquake
   b. Design horizontal acceleration coefficient
   c. Intensity of earthquake
   d. Magnitude of earthquake

(b) What are the assumptions made in the earthquake resistant design of structures as per codal provisions?

(c) What are the safety factors and load combinations considered in the ERDB?